

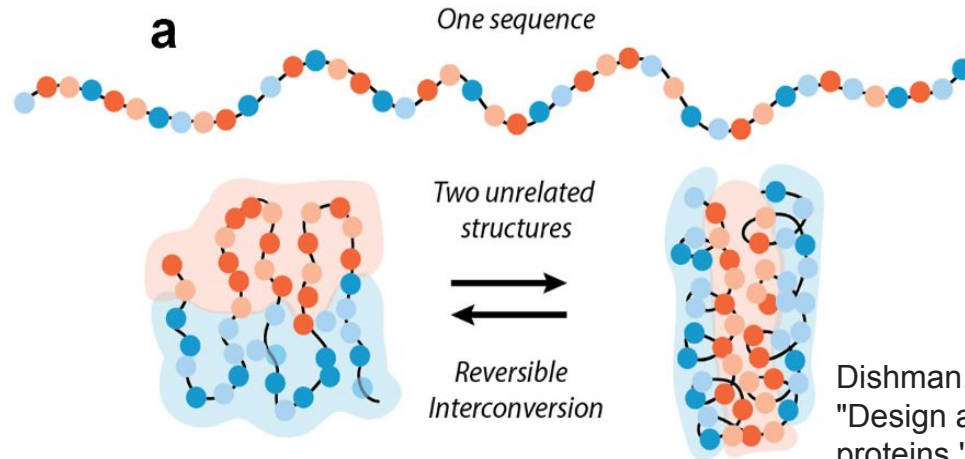
Fold Switching in Metamorphic Proteins

30th May, 2023

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Metamorphic Proteins

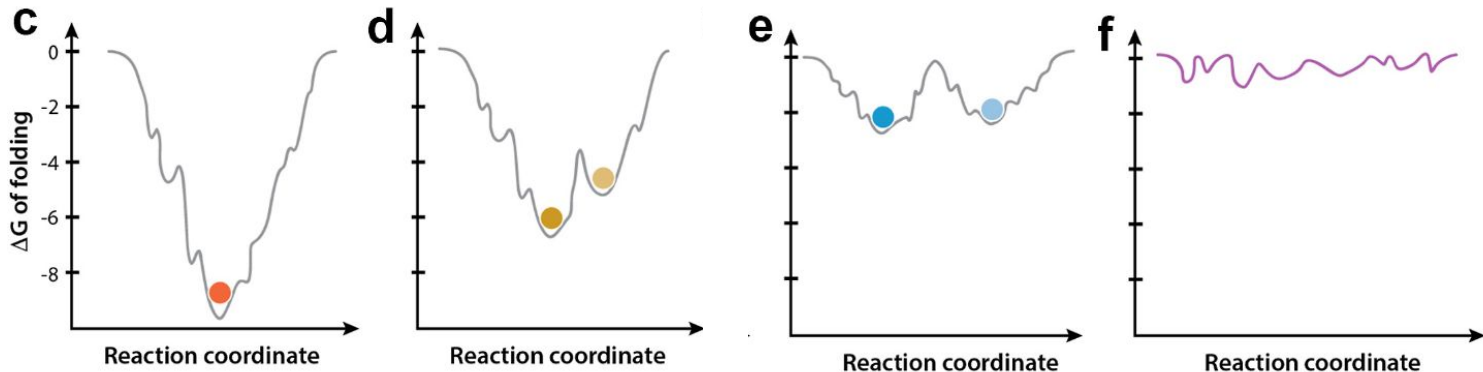
- Metamorphic proteins **defy the protein folding paradigm**, in which each amino acid sequence adopts one defined fold (monomorphic).
- They **switch reversibly between entirely distinct folds**, often with different functions, on the time scale of seconds



Dishman, Acacia F., and Brian F. Volkman. "Design and discovery of metamorphic proteins." *Current opinion in structural biology* 74 (2022): 102380.

Characteristics of a Metamorphic Protein

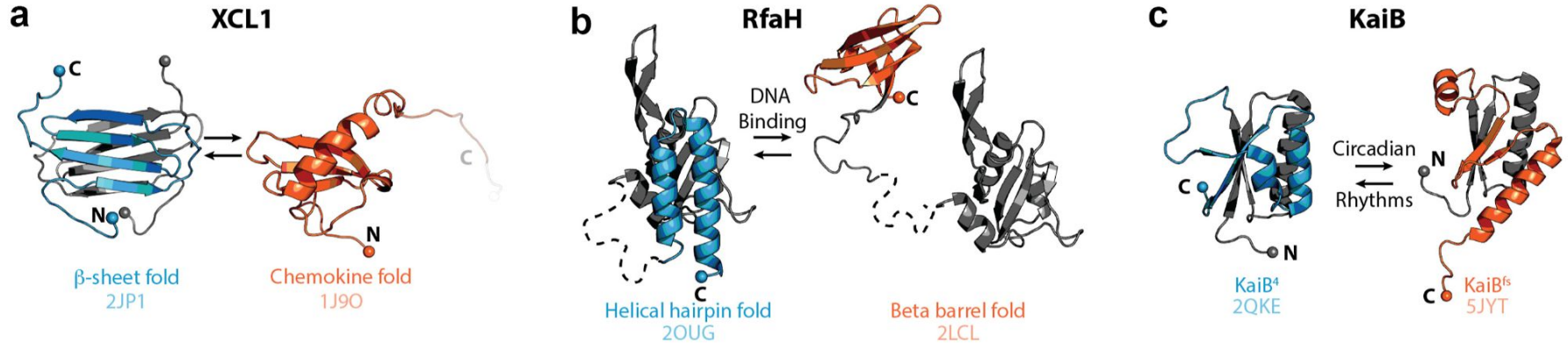
- **Marginal thermostability (i.e., $dG_{fold} > -3$ kcal/mol)**, in contrast to dG_{fold} values ranging from -15 to -5 kcal/mol for most globular proteins.
- **Spontaneous unfolding**, which makes alternative native-state structures readily accessible.
- **Structural dissimilarity** involving helix–sheet conversions or changes to β -sheet topology.



What is not a metamorphic protein?

- **Chameleon sequences**, which are identical sequence fragments encompassed by larger sequences that differ
- **Prions**, which undergo irreversible fold changes
- **Local/subtle conformational changes**, which are less dramatic than the changes undergone by metamorphic proteins
- **Coupled binding/folding of IDPs**, which converts an unfolded protein to a folded protein, as opposed to converting one folded state to another
- **Irreversible PTM-driven conformational changes** (e.g., cleavage).
- Proteins that occupy different folds as a result of sequence modifications such as **mutations, extensions, or cleavage**

Examples of Metamorphic Proteins



Current Opinion in Structural Biology

Science

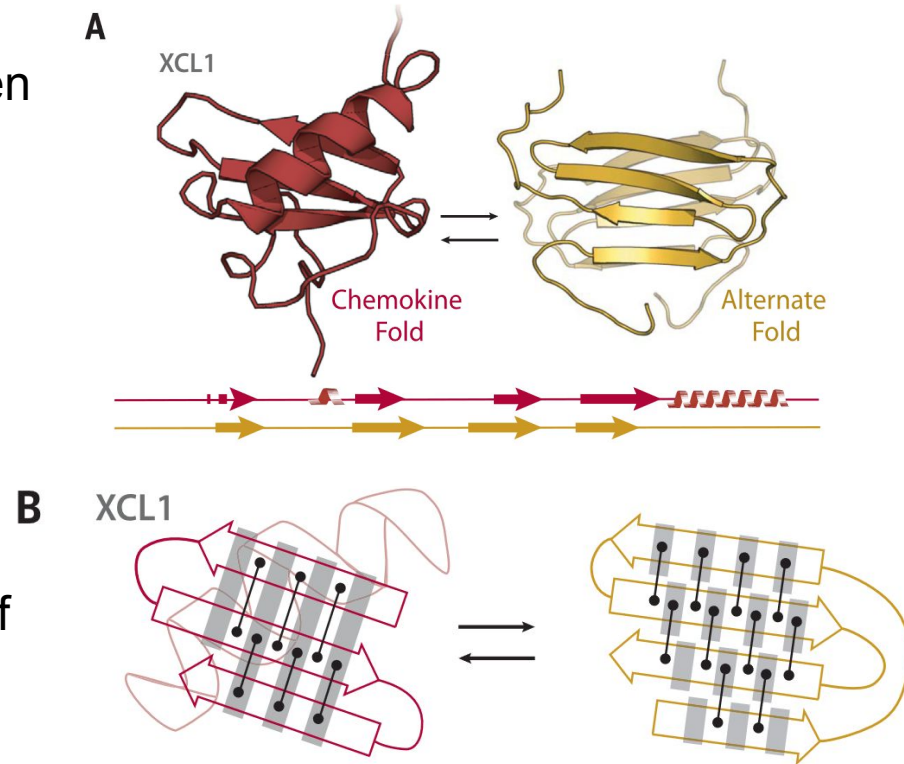
PROTEIN FOLDING

Evolution of fold switching in a metamorphic protein

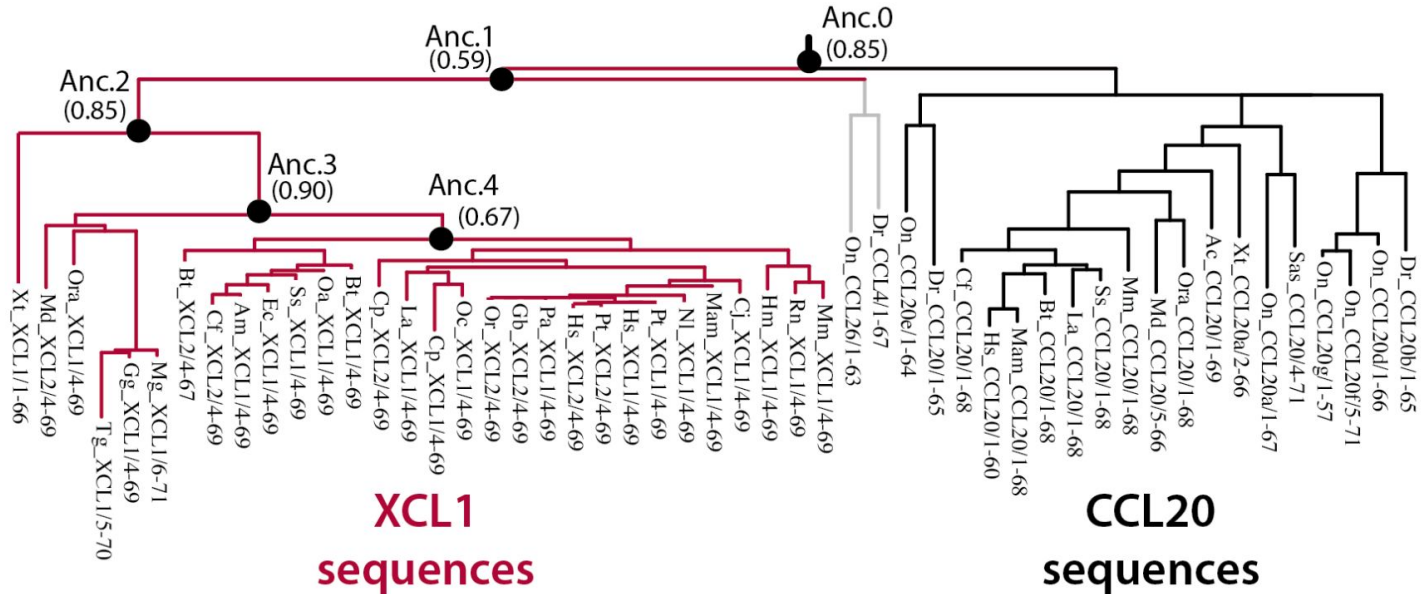
**Acacia F. Dishman^{1,2}, Robert C. Tyler¹, Jamie C. Fox¹, Andrew B. Kleist^{1,2}, Kenneth E. Prehoda³,
M. Madan Babu^{4,5}, Francis C. Peterson¹, Brian F. Volkman^{1*}**

Chemokine XCL1 (lymphotactin)

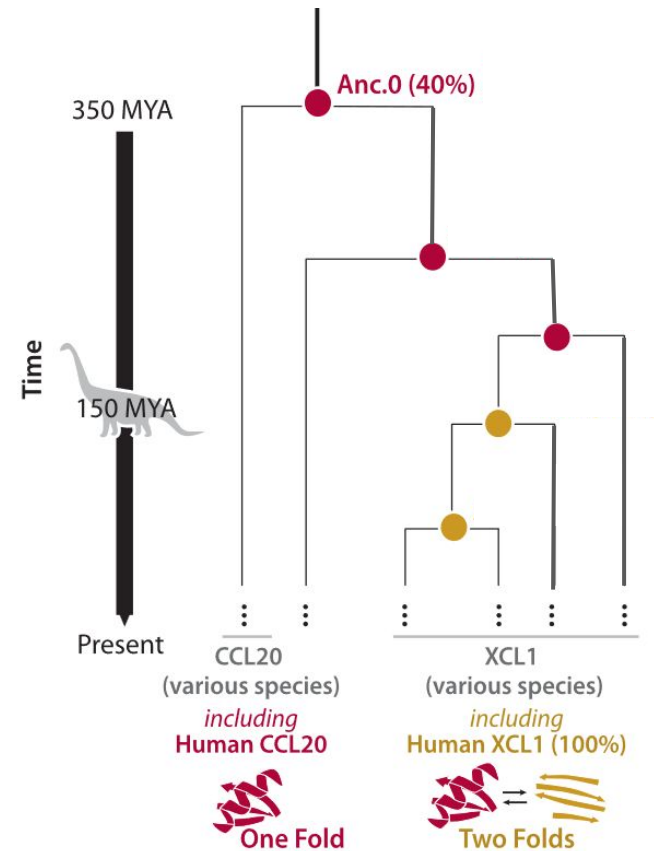
- XCL1 divides its two functions between its two folds:
 - (1) The **chemokine structure** activates XCL1's cognate **G protein-coupled receptor (GPCR)**
 - (2) The **alternate structure** binds **glycosaminoglycans (GAGs)**Additionally, the alternate structure is directly **antimicrobial**, like a subset of chemokines.



- An MSA was created for - **457 chemokine sequences** from **30 vertebrate species** incorporating **14 chemokine structures**
- Phylogenetic analysis indicated that XCL1 is most **closely related** to another (monomorphic) chemokine, **CCL20**

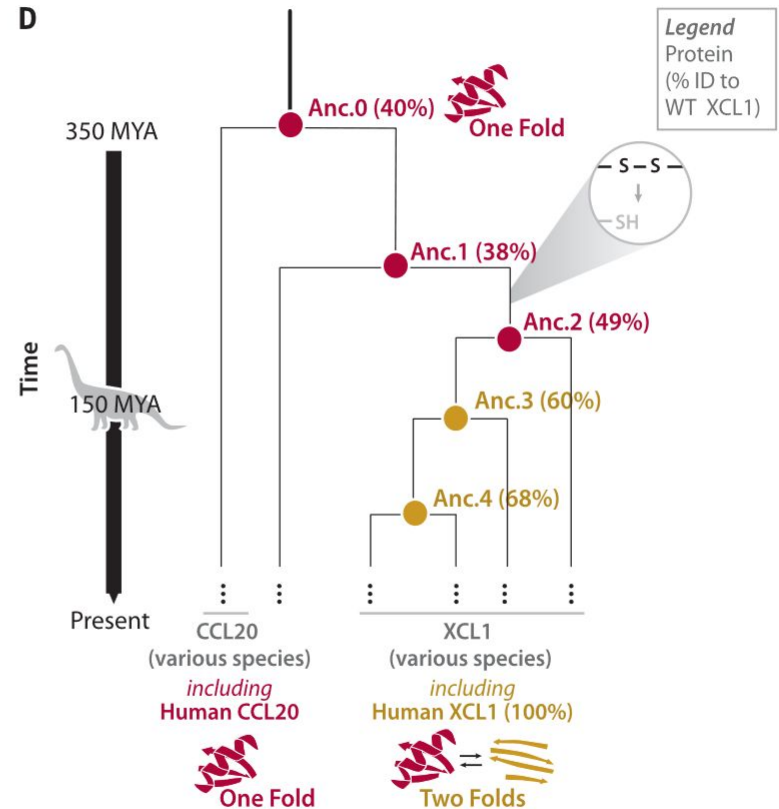


- Using **ancestral reconstruction**, the last shared ancestor of XCL1 and CCL20 (**Anc.0**) was resurrected (i.e., expressed and purified).
- Unlike the temperature- and salt-dependent structural equilibrium of XCL1, the **Anc.0 heteronuclear single-quantum coherence (HSQC) spectrum is unchanged** from 10° to 50°C, with or without NaCl
- Anc.0's structure is highly similar to human XCL1's chemokine fold (root mean square deviation of **1.45 Å**)



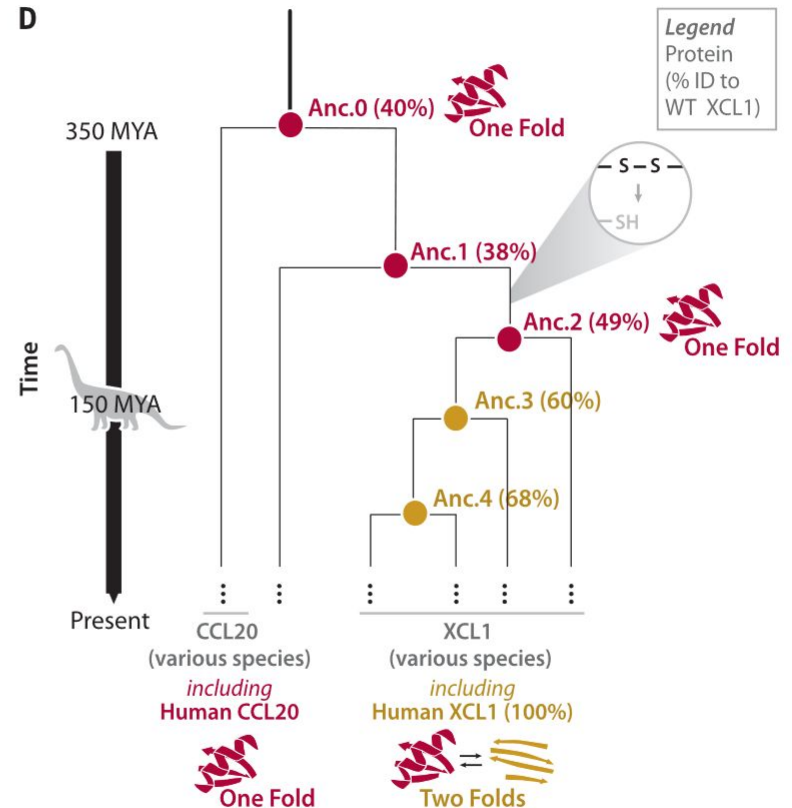
A metamorphic protein evolved from a monomorphic ancestor

- Non-XCL1 chemokines have **two conserved disulfide bonds**, one of which is incompatible with the alternate fold of XCL1.
- Locating the **oldest node whose sequence lacks the cysteines needed for one of the disulfides**, i.e **Anc. 2** , was **resurrected**.
- Anc.2 adopts a single fold** at all temperature and salt conditions tested by NMR (HSQC)

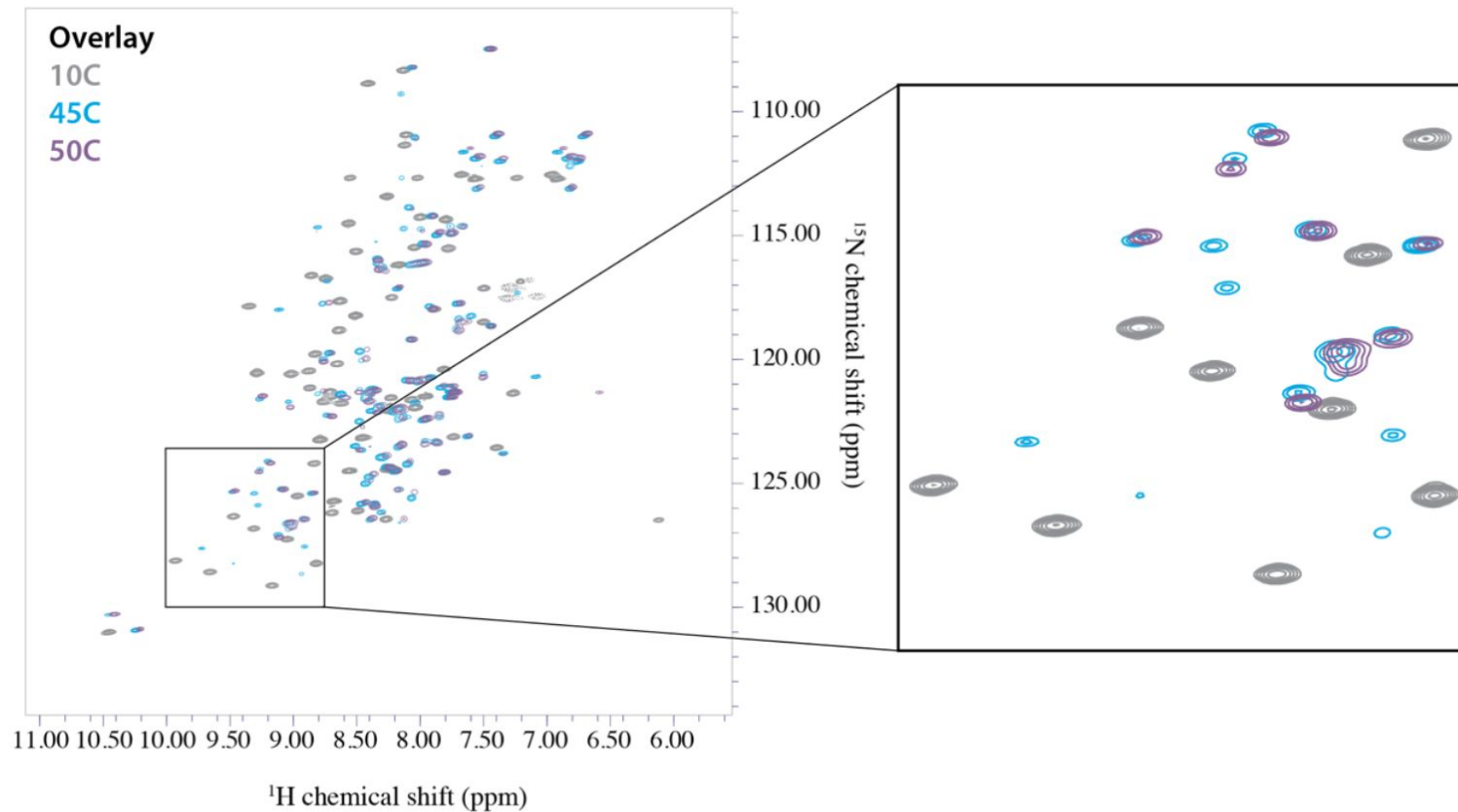


Disulfide loss alone did not enable metamorphism

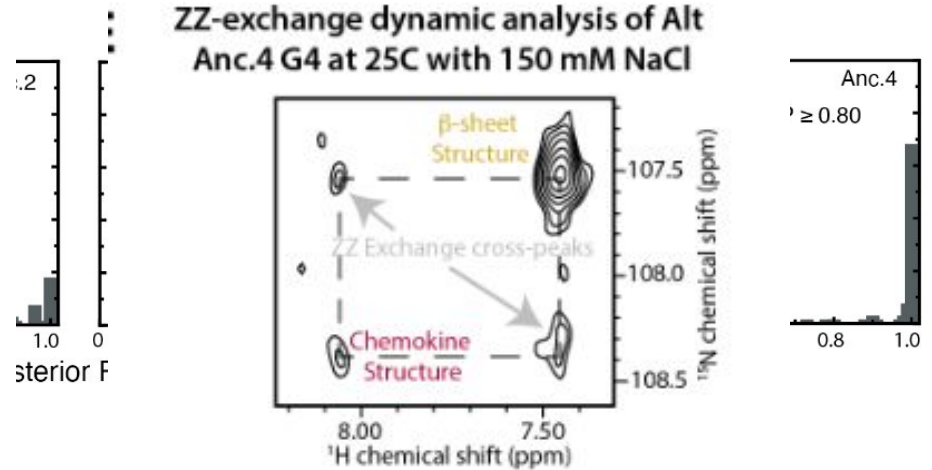
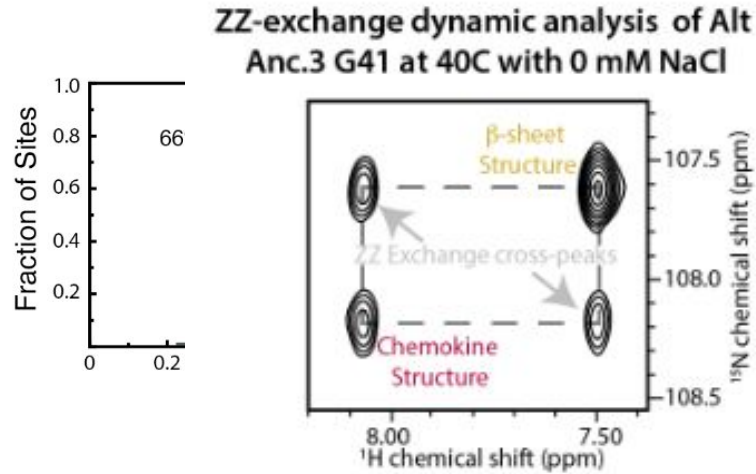
- The remaining inferred ancestral sequences (**Anc.3 and Anc.4**) were **resurrected** - NMR showed that they both interconvert reversibly between two distinct folds
- They share only 60 and 68% sequence identity with human XCL1, respectively.
- **ZZ-exchange experiments** revealed that Anc.3 [rate of exchange (kex), 0.97 s⁻¹] and Anc.4 (kex, 2.6 s⁻¹) **exchange on the time scale of seconds**, similar to human XCL1 [kex, 0.90 s⁻¹]



*** ZZ-exchange NMR experiments are used to measure the kinetics of interconversion between different molecular states**

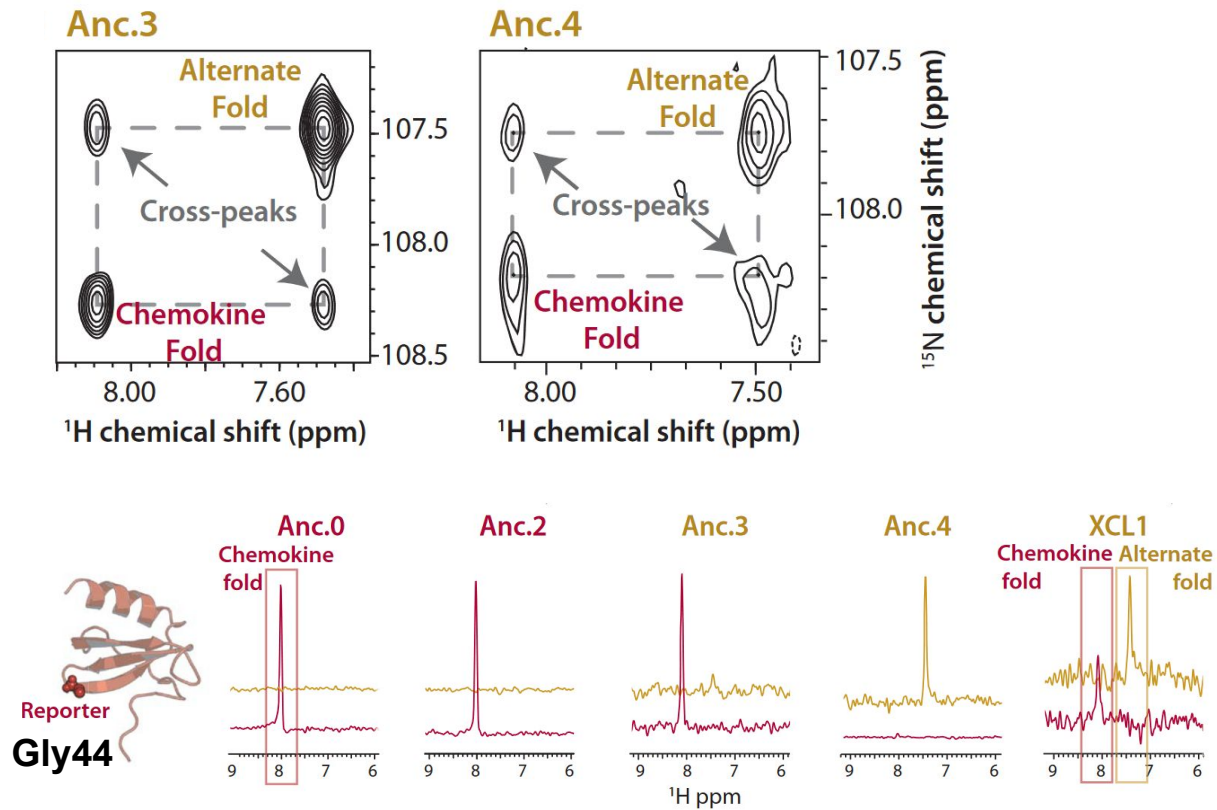


HSQC Spectra for Anc.3 at 10, 25, 37, 40, 45, and 50°C, 0 mM NaCl, 20 mM NaPO₄ pH 6.0

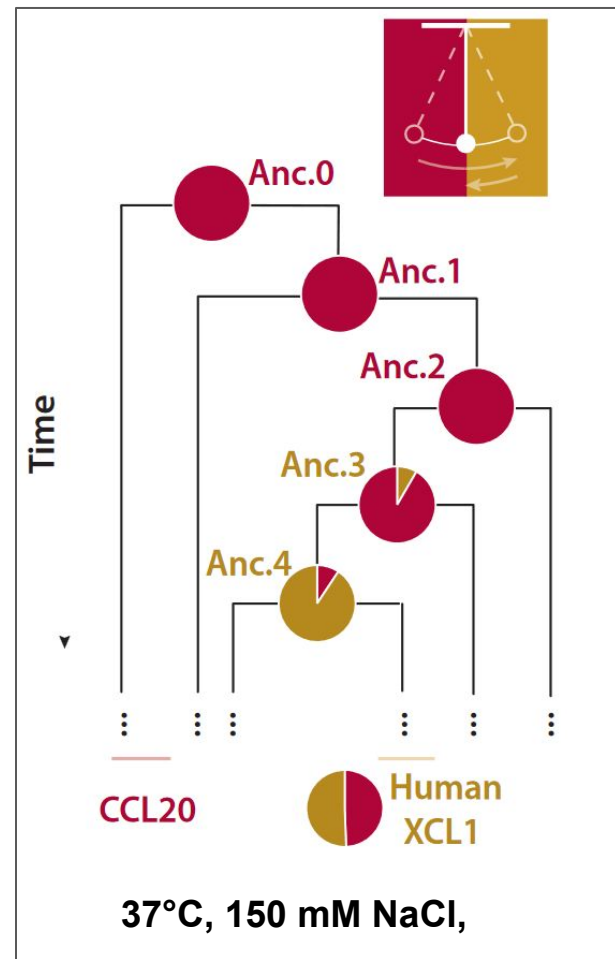


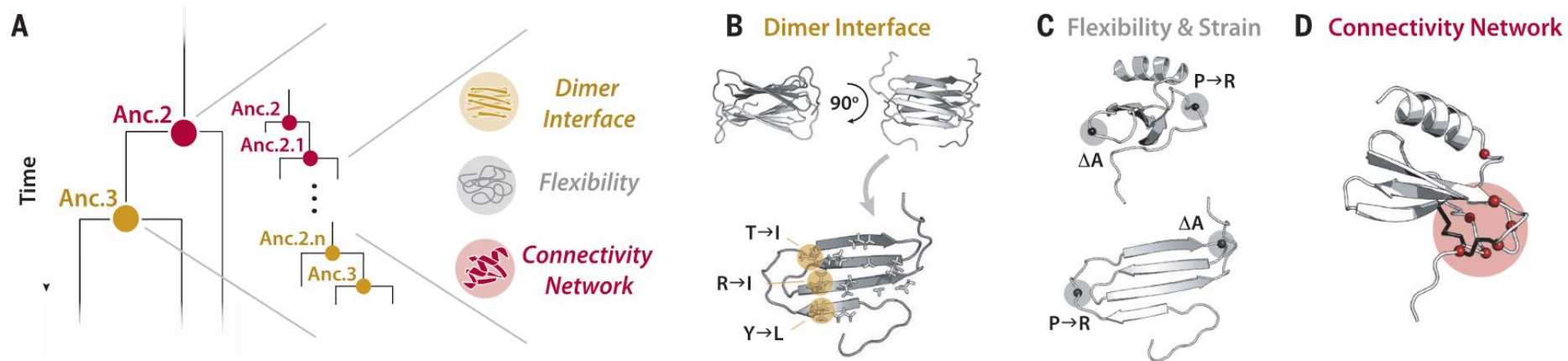
Every position at which the next most likely alternate residue was predicted with **above 20% probability** was replaced with the **second most likely amino acid**, creating a “worst case” ancestor with the alternate amino acid at all possible positions

“Alt” Ancestor Proteins Behave Similarly to Maximum Likelihood Ancestor Proteins

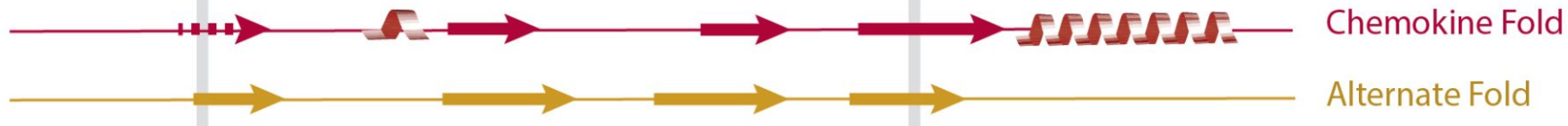


ZZ-exchange dynamic analysis of Anc.3 at 40°C and Anc.4 at 25°C, both with 0 mM NaCl, using the glycine reporter.

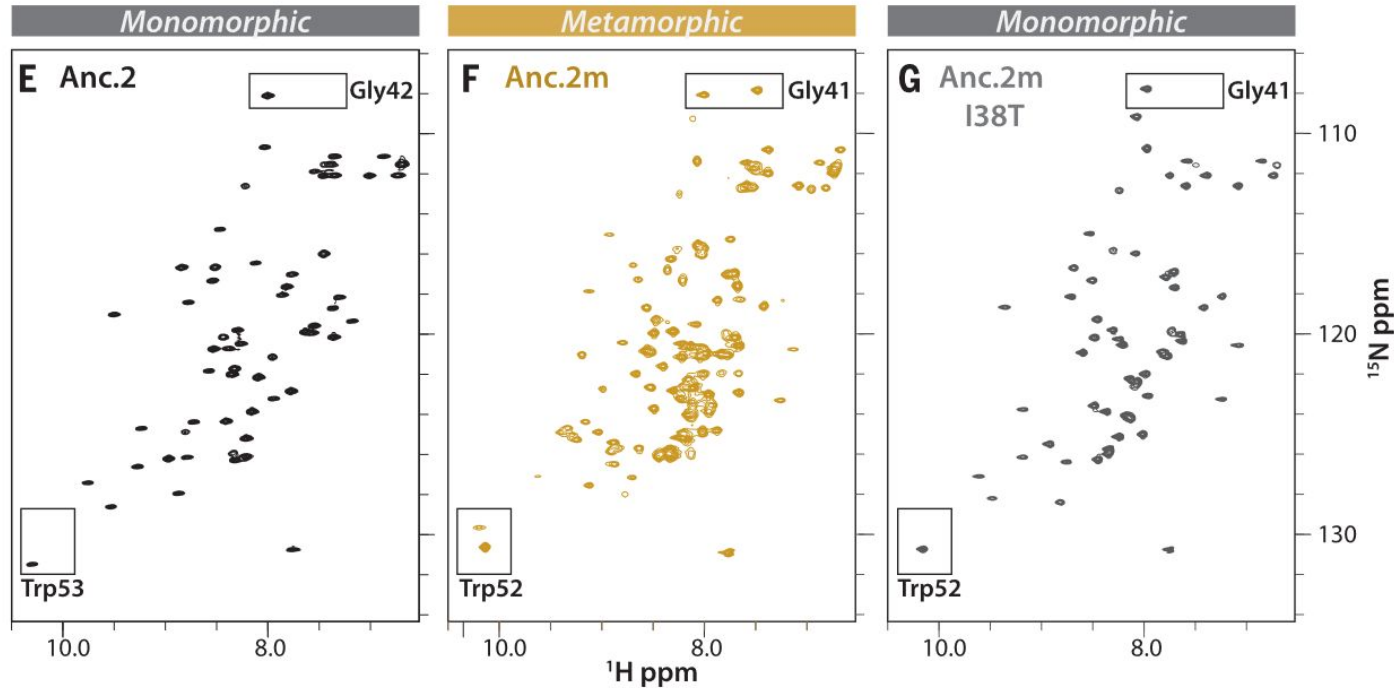




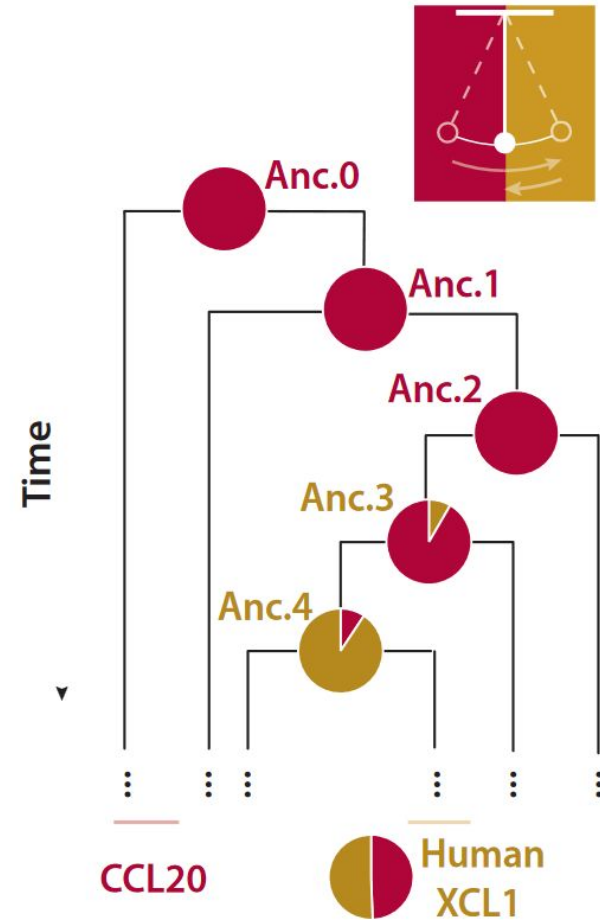
	1-3 disulfide bond	2-4 disulfide bond	%ID to XCL1
Anc.0	----ARKSCCLKYTKRPLPLKRIKSYTIQSNEAC	NIKAIIFTTKKGRKICANPNEKWVQKAMKHLDDK--	40%
Anc.2	---EVVRKSVCLKYTTQPLPLNRIKSYTIQN--	ACSIKAIIFTTKKGRKICANPEEKWVKKAIKHLDRKA--	49%
Anc.3	---EVMKKSICVSLTTQRLPINSIKSYTIQE--	GSMKAVIFITRRGIKICADPEASWVKKAIKTIDRRTTT	60%
Anc.4	---EVLEKSICVSLTTQRLPINNIKTYTIKE--	GSMKAVIFITRRGLKICADPQAKWVKKAVKTIDRKTRT	68%
XCL1	VGSEVSDKRTC VSLTTQRLPVSRIKTYTITE--	GSLRAVIFITKRGLKVCADPQATWVRDVVRSM DRKSNT	100%



Metamorphic folding is enabled by the combined presence of changes in dimer interface, structural flexibility, and residue contact networks.



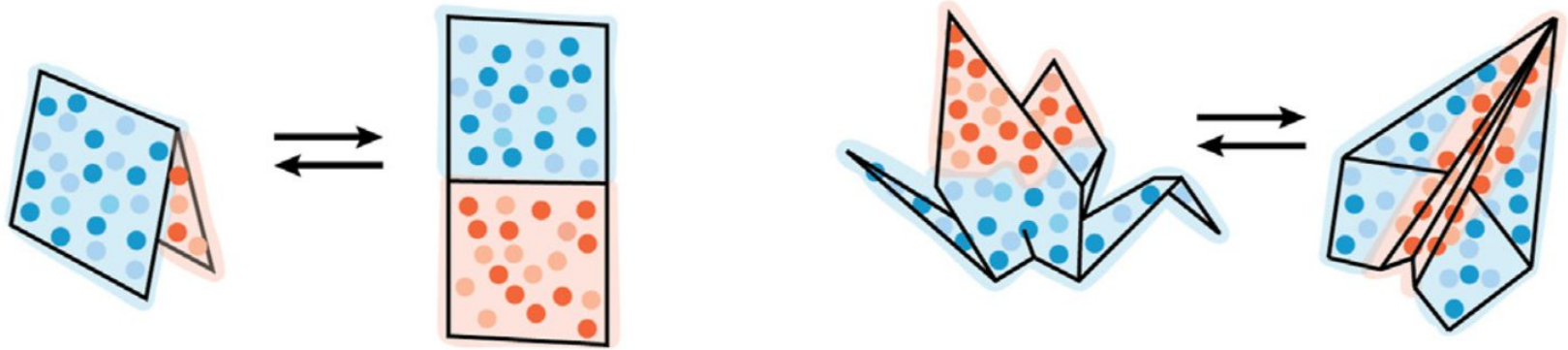
- The previously discussed positions largely cluster around **Anc.0's second disulfide bond**, “gluing” its structure in place.
- After the disulfide was lost, this remaining structural glue “dissolved” as metamorphosis evolved.
- Position 27, which points into the glue region, is a **glutamine in Anc.0, Anc.2, and Anc.3**, but in **Anc.4 it is a bulky lysine**, which is likely to sterically destabilize the chemokine fold, perhaps partly accounting for the fact that Anc.4 prefers the alternate fold.
- In **human XCL1, this position is a threonine**, perhaps facilitating XCL1's shift to a 50/50 structural equilibrium.



Conclusion

- Whereas both distributed and centralized control of folding have been described, the requirement for numerous, structurally distant mutations of different kinds to enable metamorphosis suggests that **distributed control governs XCL1's structure**.
- Metamorphosis may be a **molecular phenotype that was selected for** in XCL1 **rather than a transient feature of an evolutionary intermediate**.
- **Why would metamorphosis be favored?**
 - Metamorphic folding **could confer dynamic control** over the fractional population, and therefore the activity, of each structure.
 - To **enhance a specific function** (e.g., GAG binding) or permit acquisition of a new function (e.g., antimicrobial activity).
- Moreover, the broad span of XCL1 sequence space compatible with metamorphic interconversion implies that **design of fold-switching proteins** could be within reach, especially with the help of principles uncovered in this study

Designing Fold Switching Proteins



Directed Evolution of Highly Homologous Proteins with Different Folds by Phage Display

1. Identify conserved positions between the 2 folds
2. Introduce A binding residues in G protein
3. Random mutagenesis in specific topological islands
4. **Phage display selection** - It is based on the fact that binding of G to IgG is thermodynamically linked to its folding

Alexander, Patrick A., et al. "Directed evolution of highly homologous proteins with different folds by phage display: implications for the protein folding code." *Biochemistry* 44.43 (2005): 14045-14054.

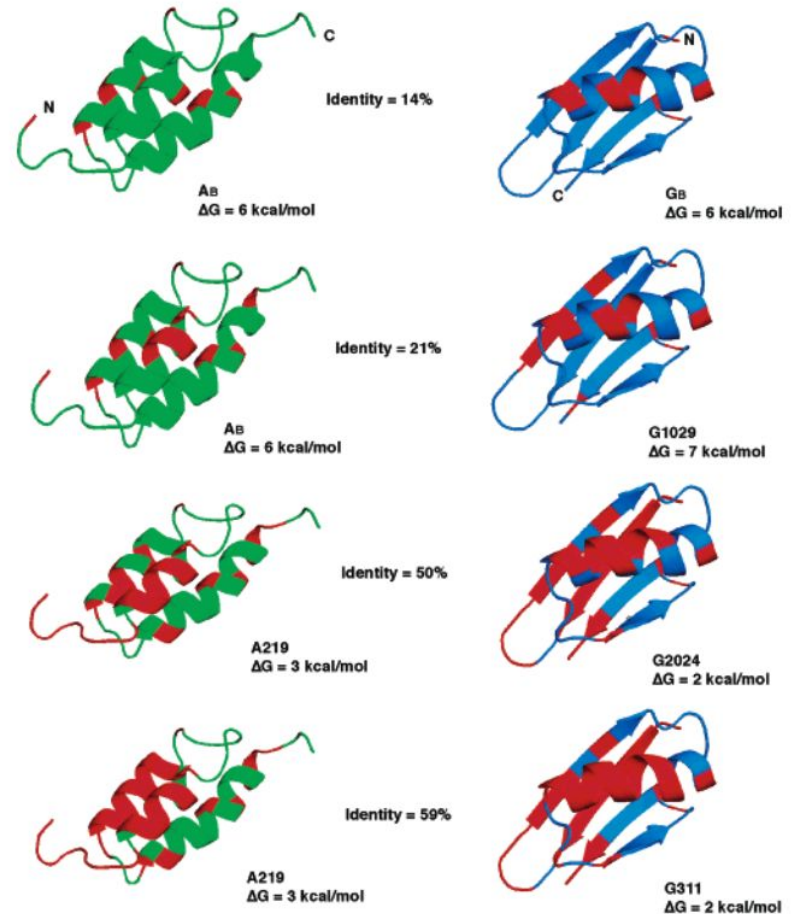
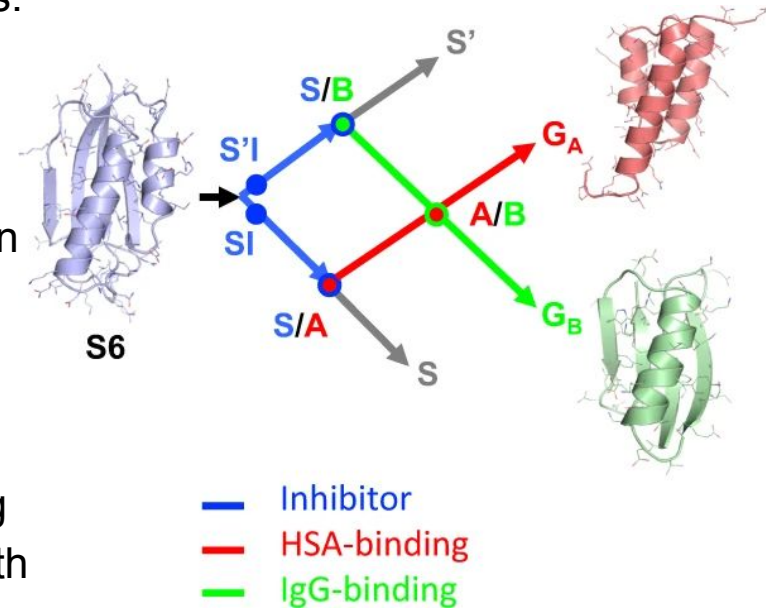


FIGURE 3: Co-evolution of homologous heteromorphs. Regions of identity between heteromorph pairs are colored red.

The design process is as follows:

- i. Thread the A- or B- sequence through both S-fold types.
- ii. Identify alignments that minimize the number of catastrophic interactions.
- iii. Design mutations to resolve unfavorable interactions in clusters of 4–6 amino acids using Pymol and energy minimize using Rosetta-Relax.
- iv. Optimize protein stability in the S-fold by computationally mutating amino acids at non-overlapping positions. Repeat energy minimization and evaluation with Rosetta-Relax.
- v. To reduce uncertainties involved in computational design, conserve original amino acids whenever possible.



Ruan, Biao, et al. "Design and characterization of a protein fold switching network." *Nature Communications* (2023)